

7.4 Integration of Rational Functions by Partial Fractions

In this section we learn to express rational functions as a sum of simpler fractions known as _____.

Recall: What are rational functions?

Motivation

How can you rewrite $f(x) = \frac{1}{x-4} - \frac{1}{x-5}$?

Now, if you think about finding an antiderivative of f , which form is easier to work with?

We often simplify expressions by writing them as a single term like you did above, but in this section our technique is the opposite!

The Method of Partial Fractions

Consider the rational function, $f(x) = \frac{h(x)}{g(x)}$.

1. The degree of the numerator must be _____ the degree of the denominator. If this condition is not met, we must use _____ to accomplish it.

2. Factor _____, $g(x)$.

3. Let $x - r$ be a linear factor of $g(x)$. Suppose $(x - r)^m$ is the highest power of $x - r$ that divides $g(x)$. To this factor, assign the sum of m partial fractions

$$\frac{A_1}{(x - r)} + \frac{A_2}{(x - r)^2} + \frac{A_3}{(x - r)^3} + \dots + \frac{A_m}{(x - r)^m}$$

4. Let $x^2 + px + q$ be an irreducible quadratic factor of $g(x)$ so that $x^2 + px + q$ has no real roots (where a check for this is that _____). Suppose that $(x^2 + px + q)^n$ is the highest power of this factor that divides $g(x)$. Then to this factor, assign the sum of the n partial fractions

$$\frac{B_1x + C_1}{(x^2 + px + q)} + \frac{B_2x + C_2}{(x^2 + px + q)^2} + \dots + \frac{B_nx + C_n}{(x^2 + px + q)^n}.$$

5. Set the original fraction equal to the sum of all partial fractions.
6. Solve for _____.
7. Integrate.

Example 1

Find the partial fraction decomposition of the following functions. You do not need to integrate or solve for the coefficients.

1.
$$f(x) = \frac{x^3 - x + 1}{(x - 1)^2(x^2 + 1)(x^2 + 4)^3}$$

2.
$$g(x) = \frac{-3x + 1}{(x^2 + 5x + 6)(x^2 + x + 3)}$$

Example 2

Evaluate $\int \frac{x^4}{x-1} dx$

When decomposing into partial fractions, how do we find the coefficients?

Example 3

Evaluate $\int \frac{-2x - 6}{x^2 - 2x - 3} dx$

Example 4

Evaluate $\int \frac{1}{x^2(2x-1)} dx$